Operating Systems Banana Problem

Part A: The Hunt

Phase 1: Deployment & Forensics 1. Compile and Run

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ./hive
Usage: ./hive <SRN>
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ./hive PES2UG24CS417
[Hive 2186] Starting. SRN=417, L=5, Z=6, M=67
[Hive 2186] Home Directory: /home/bhava
[Hive 2186] Current Directory: /mnt/c/Users/bhava/OneDrive/Desktop/OS
[Hive 2186] Spawned 5 larvae and 6 husks.
[Hive 2186] Control commands: type 'reap' to reap zombies, 'quit' to shutdown.
[Hive 2186] >
```

2. Locate the Target

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ cat/tmp/hive.pid
bash: cat/tmp/hive.pid: No such file or directory
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ find /tmp -name "*hive*"
find: ‘/tmp/systemd-private-984e64ea253e4039a3632e2ddc9a903b-systemd-logind.service-Yd3JHj’: Permission denied
find: ‘/tmp/snap-private-tmp’: Permission denied
/tmp/lab_417/hive.pid
find: ‘/tmp/systemd-private-984e64ea253e4039a3632e2ddc9a903b-wsl-pro.service-TFpBSA’: Permission denied
find: ‘/tmp/systemd-private-984e64ea253e4039a3632e2ddc9a903b-systemd-resolved.service-I4eLlu’: Permission denied
find: ‘/tmp/systemd-private-984e64ea253e4039a3632e2ddc9a903b-systemd-timesyncd.service-4mzLMk’: Permission denied
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ cat /tmp/lab_417/hive.pid
2186
```

3. Map the Family Tree

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -ef | grep hive | grep -v grep
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$
```

4. Isolate the Undead

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -o pid,ppid,state,cmd --ppid 417
    PID    PPID S CMD
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -eo pid,ppid,state,cmd | awk '$3=="Z"'
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$
```

5. Analyze Memory Consumption

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -eo pid,ppid,state,cmd --ppid 417
  PID    PPID S CMD
    1       0 S /sbin/init
    2       1 S /init
    6       2 S plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
   42       1 S /usr/lib/systemd/systemd-journald
   89       1 S /usr/lib/systemd/systemd-udevd
  104       1 S /usr/lib/systemd/systemd-resolved
  105       1 S /usr/lib/systemd/systemd-timesyncd
  168       1 S /usr/sbin/cron -f -P
  169       1 S @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
  182       1 S /usr/lib/systemd/systemd-logind
  190       1 S /usr/libexec/wsl-pro-service -vv
  196       1 S /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
  216       1 S /sbin/agetty -o -p -- \u --noclear - linux
  219       1 S /usr/sbin/rsyslogd -n -iNONE
  247       1 S /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
  358       2 S /bin/login -f
  402       1 S /usr/lib/systemd/systemd --user
  403     402 S (sd-pam)
  423     358 S -bash
  512       2 S /init
  513     512 S /init
  514     513 S sh -c "$VSCODE_WSL_EXT_LOCATION/scripts/wslServer.sh" c9d77990917f3102ada88be140d28b038d1dd7c7 stable code-server .vscode-server --host=127.0.0.
  515     514 S sh /mnt/c/Users/bhava/.vscode/extensions/ms-vscode-remote.remote-wsl-0.104.3/scripts/wslServer.sh c9d77990917f3102ada88be140d28b038d1dd7c7 stabl
  521     515 S sh /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/bin/code-server --host=127.0.0.1 --port=0 --connection-token=98796429
  525     521 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1d
  536       2 S /init
  537     536 S /init
  538     537 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node -e const net = require('net'); process.stdin.pause(); const client
  545     525 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1d
  556       2 S /init
  557     556 S /init
  562     557 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node -e const net = require('net'); process.stdin.pause(); const client
  578     525 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1d
  597     525 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node --dns-result-order=ipv4first /home/bhava/.vscode-server/bin/c9d7799
 2733     545 S /bin/bash --init-file /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/out/vs/workbench/contrib/terminal/common/scripts/s
 8435    2733 R ps -eo pid,ppid,state,cmd --ppid 417
```

6. Investigate Logs

```
[Larva 2191] Allocated 1 MB (Total: 27 MB)
[Larva 2191] Allocated 1 MB (Total: 28 MB)
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[Larva 2191] Allocated 1 MB (Total: 63 MB)
[Larva 2191] Allocated 1 MB (Total: 64 MB)
[Larva 2191] malloc failed at 65 MB; going dormant
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[Larva 2188] Allocated 1 MB (Total: 53 MB)
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[Larva 2188] Allocated 1 MB (Total: 64 MB)
[Larva 2188] malloc failed at 65 MB; going dormant
[Larva 2189] started (M=67 MB)
[Larva 2189] Allocated 1 MB (Total: 1 MB)
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[Larva 2187] Allocated 1 MB (Total: 62 MB)
[Larva 2187] Allocated 1 MB (Total: 63 MB)
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[Larva 2187] malloc failed at 65 MB; going dor
[Larva 2188] started (M=67 MB)
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[Larva 2187] Allocated 1 MB (Total: 53 MB)
```

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -eo pid,ppid,state,cmd --ppid 417
    PID    PPID S CMD
      1       0 S /sbin/init
      2       1 S /init
      6       2 S plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
     42       1 S /usr/lib/systemd/systemd-journald
     89       1 S /usr/lib/systemd/systemd-udevd
    104       1 S /usr/lib/systemd/systemd-resolved
    105       1 S /usr/lib/systemd/systemd-timesyncd
    168       1 S /usr/sbin/cron -f -P
    169       1 S @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
    182       1 S /usr/lib/systemd/systemd-logind
    190       1 S /usr/libexec/wsl-pro-service -vv
    196       1 S /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
    216       1 S /sbin/agetty -o -p -- \u --noclear - linux
    219       1 S /usr/sbin/rsyslogd -n -iNONE
    247       1 S /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
    358       2 S /bin/login -f
    402       1 S /usr/lib/systemd/systemd --user
    403     402 S (sd-pam)
    423     358 S -bash
    512       2 S /init
    513     512 S /init
    514     513 S sh -c "$VSCODE_WSL_EXT_LOCATION/scripts/wslServer.sh" c9d77990917f3102ada88be140d28b038d1dd7c7 stable code-server .vscode-server --host=127.0.0.
    515     514 S sh /mnt/c/Users/bhava/.vscode/extensions/ms-vscode-remote.remote-wsl-0.104.3/scripts/wslServer.sh c9d77990917f3102ada88be140d28b038d1dd7c7 stabl
    521     515 S sh /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/bin/code-server --host=127.0.0.1 --port=0 --connection-token=98796429
    525     521 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1d
    536       2 S /init
    537     536 S /init
    538     537 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node -e const net = require('net'); process.stdin.pause(); const client
    545     525 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1d
    556       2 S /init
    557     556 S /init
    562     557 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node -e const net = require('net'); process.stdin.pause(); const client
    578     525 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1d
    597     525 S /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/node --dns-result-order=ipv4first /home/bhava/.vscode-server/bin/c9d7799
   2733     545 S /bin/bash --init-file /home/bhava/.vscode-server/bin/c9d77990917f3102ada88be140d28b038d1dd7c7/out/vs/workbench/contrib/terminal/common/scripts/s
   8435    2733 R ps -eo pid,ppid,state,cmd --ppid 417
```

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ cat /tmp/lab_417/larve_417.log
cat: /tmp/lab_417/larve_417.log: No such file or directory
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ cat /tmp/lab_417/larva_*.log
[Larva 2187] started (M=67 MB)
[Larva 2187] Allocated 1 MB (Total: 1 MB)
[Larva 2187] Allocated 1 MB (Total: 2 MB)
[Larva 2187] Allocated 1 MB (Total: 3 MB)
[Larva 2187] Allocated 1 MB (Total: 4 MB)
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[Larva 2187] Allocated 1 MB (Total: 8 MB)
[Larva 2187] Allocated 1 MB (Total: 9 MB)
[Larva 2187] Allocated 1 MB (Total: 10 MB)
[Larva 2187] Allocated 1 MB (Total: 11 MB)
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```

7. Verify Parent/Child Links (PID/PPID)

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ pstree -p 417
```

Phase 2: The Cure

1. Reap the Zombies

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ./hive PES2UG24CS417
[Hive 10001] Starting. SRN=417, L=5, Z=6, M=67
[Hive 10001] Home Directory: /home/bhava
[Hive 10001] Current Directory: /mnt/c/Users/bhava/OneDrive/Desktop/OS
[Hive 10001] Spawned 5 larvae and 6 husks.
[Hive 10001] Control commands: type 'reap' to reap zombies, 'quit' to shutdown.
[Hive 10001] >
[Hive 10001] >
[Hive 10001] Unknown command '  '. Try: reap | quit
[Hive 10001] >
[Hive 10001] > REAP
[Hive 10001] Unknown command 'REAP'. Try: reap | quit
[Hive 10001] > reap
[Hive 10001] Command received: reap (reaping zombies)...
[Hive 10001] Reaped 6 zombie(s).
[Hive 10001] > quit
[Hive 10001] Command received: quit (shutting down)...
[Hive 10001] Shutdown complete.
```

2. Verify Cleanup

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -e -o pid,ppid,state,cmd |grep 417
  10891    2733 S grep --color=auto 417
```

3. Check the Larvae

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ps -e -o pid,ppid,state,cmd |grep 417
  10891    2733 S grep --color=auto 417
```

Phase 3: Sterilization

```
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$ ./hive PES2UG24CS417
[Hive 11027] Starting. SRN=417, L=5, Z=6, M=67
[Hive 11027] Home Directory: /home/bhava
[Hive 11027] Current Directory: /mnt/c/Users/bhava/OneDrive/Desktop/OS
[Hive 11027] Spawned 5 larvae and 6 husks.
[Hive 11027] Control commands: type 'reap' to reap zombies, 'quit' to shutdown.
[Hive 11027] > reap
[Hive 11027] Command received: reap (reaping zombies)...
[Hive 11027] Reaped 6 zombie(s).
[Hive 11027] > quit
[Hive 11027] Command received: quit (shutting down)...
[Hive 11027] Shutdown complete.
bhava@LAPTOP-4NR2G1UV:/mnt/c/Users/bhava/OneDrive/Desktop/OS$
```

Part B: Code Analysis (Theory)

1. How can you verify the parent/child relationship between the Hive and its children?

PID is the process ID.PPID is the parent's process IDIf a process has PPID equal to the Hive's PID, then it is a child of the Hive.

2. Why do zombies exist until the parent reaps them?

When a child exits, it sends an exit status to the parent.Until the parent collects this status using wait() or waitpid(), the OS keeps the child entry.This exited but not cleaned process is called a zombie.

3.Explain the lifecycle of the "Husk" processes. Why do they become zombies?

usk processes exit immediately after fork().The parent (Hive) does not call wait() immediately.So the husks stay in the process table as zombiesThey are removed only when the reap command is given

4. Why does a Larva use while(1) pause(); instead of exit()?

This keeps the larva alive.The allocated memory is not releaseThe process stays in sleeping state instead of terminating.

5. What happens if all larvae exit while the parent is sleeping?

All larvae would terminate. Memory would be freed .The Hive would not consume resources anymoreThis defeats the purpose of the program.